

IKHSANNUDIN LATHIEF

Data Engineer

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PROFESSIONAL SUMMARY

Informatics graduate specializing in Data Engineering with practical expertise in designing scalable ETL/ELT pipelines, dimensional data models, and modern data warehouse architectures. Experienced in implementing idempotent workflow orchestration, automated data quality validations, and query performance optimizations. Dedicated to delivering reliable, analytics-ready models to empower data-driven decision-making, complemented by practical software engineering experience in real-time data streaming and system integration.

EDUCATION

Institut Teknologi Sumatera

Bachelor of Informatics Engineering (S.Kom.) | GPA: 3.86 / 4.00

Lampung, Indonesia
August 2022 – July 2026

- **Final-Year Project:** Application of Curriculum Learning Technique in IndoBERT-lite Model Training for Sarcasm Classification in Natural Indonesian Text
- **Relevant Subjects:** Database Basics, Database Management, Cloud Computing, Machine Learning, Object Oriented Programming

SKILLS & LANGUAGES

Core Concepts: ETL/ELT, Data Modeling, Medallion Architecture, Data Warehousing, Slowly Changing Dimensions (SCD), SQL Optimization

Data Processing & Orchestration: Apache Airflow, Apache Spark (PySpark), dbt, Pandas, NumPy

Programming & Databases: Python, SQL, PostgreSQL

Cloud & Warehousing: AWS (S3, Glue, Redshift), GCP (Cloud Storage, BigQuery)

Development Tools: Docker, Git, Linux, Jupyter

Languages: Indonesian (Native), English (Basic to Intermediate)

DATA ENGINEERING & RESEARCH PROJECTS

Taxi Trip Data Pipeline | ETL, AWS (S3, Glue, Redshift), Spark (PySpark), Apache Airflow, Streamlit 2026

- Processed 44M+ taxi trip records through an end-to-end ETL pipeline following the Medallion architecture (Bronze, Silver, Gold) in Amazon S3 using PySpark (Core API & Spark SQL) on AWS Glue.
- Designed a Star Schema (1 fact, 5 dimensions) in Amazon Redshift, optimizing query performance and RPU costs via SORTKEY and DISTKEY strategies (zone map pruning & colocated joins).
- Orchestrated idempotent pipelines and safe historical backfills in Apache Airflow via atomic partition replacements, connecting curated data to an interactive Streamlit BI dashboard via Redshift Data API.

E-Commerce Data Pipeline | ETL, Spark (PySpark), AWS S3, Apache Airflow, Streamlit 2026

- Processed and cleaned 100K+ e-commerce orders across 9 relational tables using PySpark and AWS S3 in an end-to-end batch data pipeline.
- Designed a Star Schema with 5 dimensions and 3 fact tables, implementing SCD Type 2 to track customer relocation history over time.
- Orchestrated a 9-job Airflow pipeline with dynamic partition overwrites for idempotent backfills, validating data contracts with Pytest before publishing 4 business marts to a Streamlit dashboard.

Wikipedia API Data Pipeline | ELT, Apache Airflow, dbt, BigQuery 2026

- Developed an ELT pipeline processing 5,000+ Wikipedia records, orchestrating Airflow to ingest API data into Google Cloud Storage and load raw tables into BigQuery.
- Engineered modular data transformations across staging and marts layers using dbt and SQL in BigQuery, implementing automated data tests to detect anomalies and ensure integrity.
- Optimized BigQuery query performance and storage cost via partitioning and clustering, utilizing Airflow backfill capabilities to idempotently reprocess historical data without duplication.

Indonesian Sarcasm Classification | Pandas, NumPy, Matplotlib, Jupyter, PyTorch 2026

- Conducted undergraduate thesis research applying curriculum learning with IndoBERT-lite, building an end-to-end text preprocessing pipeline using Python, Pandas, and Regex for noisy Reddit and Twitter text.
- Implemented dynamic training batch scheduling based on token matrix norms using NumPy and PyTorch, addressing class imbalance via dynamic class weighting and stratified splits.
- Organized data workflows and experiment tracking across 15+ modular notebooks using Pandas DataFrames for reproducible evaluation and benchmarking.

WORK EXPERIENCE

PT. Init Visual Experia
Software Engineer Intern | Project: Invix (IoT Data Pipeline & Geospatial Visualization)

Lampung, Indonesia
July 2025 – August 2025

- Prevented live tracking data loss across multiple IoT devices streaming at 1-second intervals by implementing a dual-channel WebSocket push and HTTP polling fallback.
- Improved data consistency and reduced visualization errors by building an automated transformation pipeline for parsing string numerics, mapping booleans to categorical values, and merging incremental state.
- Built an automated processing engine to transform raw location logs into structured trip timelines based on engine state transitions and driving duration.
- Developed interactive geospatial visualizations using Leaflet, OpenStreetMap reverse geocoding, route polylines, and in-memory caching.

UPA TIK ITERA
Computer and Software Practicum Assistant

Lampung, Indonesia
January 2024 – April 2024

- Provided technical guidance to 60+ undergraduate students during hands-on programming and software practicum sessions.
- Evaluated weekly programming assignments and provided detailed technical feedback to help students optimize code implementations.
- Supported lecturers in preparing and delivering practicum laboratory activities, environment setups, and troubleshooting technical issues.

CERTIFICATIONS

Data Engineer Associate | *DataCamp* (Credential ID: DEA0017615804198)

September 2026 – September 2028

SQL Associate | *DataCamp* (Credential ID: SQA0012471283289)

September 2026 – September 2028

LEADERSHIP & ORGANIZATIONS

UKM Petanque ITERA
Head of Public Relations

Lampung, Indonesia
November 2024 – August 2025

- Coordinated external communications and collaboration for joint training events, facilitating participation and engagement with athletes and organizations outside ITERA.

Head of the Athlete Development Division

November 2023 – August 2024

- Led the development of 20+ athletes through structured training and competition preparation, contributing to a national-level championship achievement.

Kumabat Cup Petanque Open Tournament (10th Dies Natalis ITERA)
Field Coordinator

Lampung, Indonesia
October 2024

- Coordinated on-site operations for a national-level Petanque tournament involving teams from multiple provinces, leading field committee members and ensuring smooth match execution.

Petanque Competition (Dies Natalis ITERA)
Head of Field Operations

Lampung, Indonesia
October 2023

- Directed field operations and match logistics for the Petanque competition during ITERA’s annual anniversary event, overseeing committee workflows and participant scheduling.